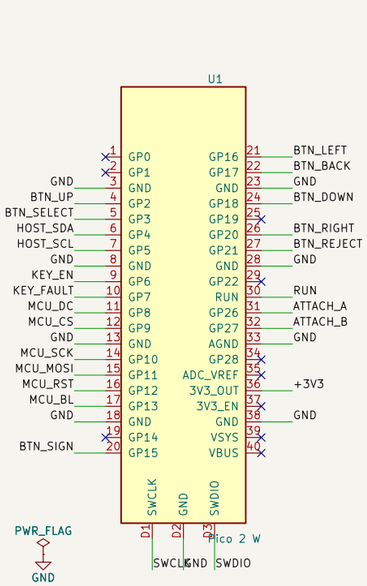
M1 CONSOLE CARRIER — USB POWER ON PICO
Eight active-low controls. Separate key protection adapter required.
No external USB-C key connector or switched key supply on this PCB.



PWR_FLAG
GND

+3V3
R1
10k
BTN_UP
UP
BTN_UP
SW1
GND
TL3305AF160QG

+3V3
R2
10k
BTN_DOWN
DOWN
BTN_DOWN
SW2
GND
TL3305AF160QG

+3V3
R3
10k
BTN_LEFT
LEFT
BTN_LEFT
SW3
GND
TL3305AF160QG

+3V3
R4
10k
BTN_RIGHT
RIGHT
BTN_RIGHT
SW4
GND
TL3305AF160QG

+3V3
R5
10k
BTN_SELECT
SELECT
BTN_SELECT
SW5
GND
TL3305AF160QG

+3V3
R6
10k
BTN_BACK
BACK
BTN_BACK
SW6
GND
TL3305AF160QG

+3V3
R7
10k
BTN_SIGN
SIGN
BTN_SIGN
SW7
GND
TL3305AF160QG

+3V3
R8
10k
BTN_REJECT
REJECT
BTN_REJECT
SW8
GND
TL3305AF160QG

SWCLK
1
TP1
SWCLK

SWDIO
1
TP2
SWDIO

GND
1
TP3
GND

RUN
1
TP4
RUN

J2
GND
+3V3
HOST_SDA
HOST_SCL
KEY_EN
KEY_FAULT
ATTACH_A
ATTACH_B
1
2
3
4
5
6
7
8
GND
+3V3
HOST_SDA
HOST_SCL
KEY_EN
KEY_FAULT
ATTACH_A
ATTACH_B
INTERNAL ADAPTER PADS

J1
+3V3
GND
LCD_MOSI
LCD_SCK
LCD_CS
LCD_DC
LCD_RST
LCD_BL
1
2
3
4
5
6
7
8
+3V3
GND
LCD_MOSI
LCD_SCK
LCD_CS
LCD_DC
LCD_RST
LCD_BL
S8B-PH-SM4-TB

MCU_MOSI
R9
33R
LCD_MOSI

MCU_SCK
R10
33R
LCD_SCK

MCU_CS
R11
33R
LCD_CS

+3V3
C1
100n
GND

MCU_DC
R12
33R
LCD_DC

MCU_RST
R13
33R
LCD_RST

MCU_BL
R14
33R
LCD_BL

+3V3
C2
10u
GND

J2: HOST PADS ONLY. Adapter must switch/limit/reverse-block key power, isolate bus, protect sense paths and handle removal. KEY_EN has a 100k default-off pull-down. LCD cable: meter every conductor to module labels before power. 3V3 operation only. No firmware, provisioning or key-lock changes. Reduce SPI clock for cable bring-up.

HOST_SDA
R15
2.2k
+3V3

HOST_SCL
R16
2.2k
+3V3

KEY_EN
R17
100k
GND

KEY_FAULT
R18
10k
+3V3

Console carrier only; protected key adapter separate; see README.md

PicoQuorum

Sheet: /
File: m1.kicad_sch

Title: PicoQuorum M1 prototype console carrier

Size: A3	Date: 2026-10-07	**Rev: M1-draft**
KiCad E.D.A. 10.0.6		Id: 1/1